

3D Printed Wind Turbine



INTRODUCTION

Electricity is essential in daily life, yet rural areas often face power shortages that disrupt basic activities. In many villages, power outages last several days, affecting students during exams and making it difficult to perform everyday tasks. Similarly, travelers and campers face problems charging devices and lighting LEDs in remote areas. To overcome these challenges, this project presents a Portable 3D Printed Wind Turbine. It harnesses wind energy and converts it into electrical power to operate small electronic devices. This lightweight and affordable design provides an efficient and eco-friendly solution suitable for rural homes, students, and outdoor activities. The turbine promotes renewable energy use and engineering innovation.



OBJECTIVE

- To design a 3D model of a small-scale functional wind turbine.
- To create airfoil-shaped blades that maximize lift and efficiency.
- To design a central hub and nacelle to house the generator and electrical components.
- To 3D print and assemble the complete turbine using lightweight and durable materials.
- To evaluate the performance and sustainability aspects of the design.



Components of wind turbine



Functions Of Components

- **Foundation**:- Foundation give support to wind turbine. It consist of solid concrete assembly
- **Tower**:- It connect rotor and foundation and raise rotor so that it can operate at required wind. Made up of steel having tube like structure.
- **Rotor & Rotor blades**: It is rotating part which converts wind energy to rotation. It is made up of glass-fibre, carbon-fibre, reinforced plastics.
- **Nacelle** :-Contains the key components of the wind turbine including the gearbox, yaw system and electric generator
- **Low speed shaft**:-Connects the rotor hub to the gearbox.
- **High speed shaft** :-Drives the electrical generator by rotating at high speed.



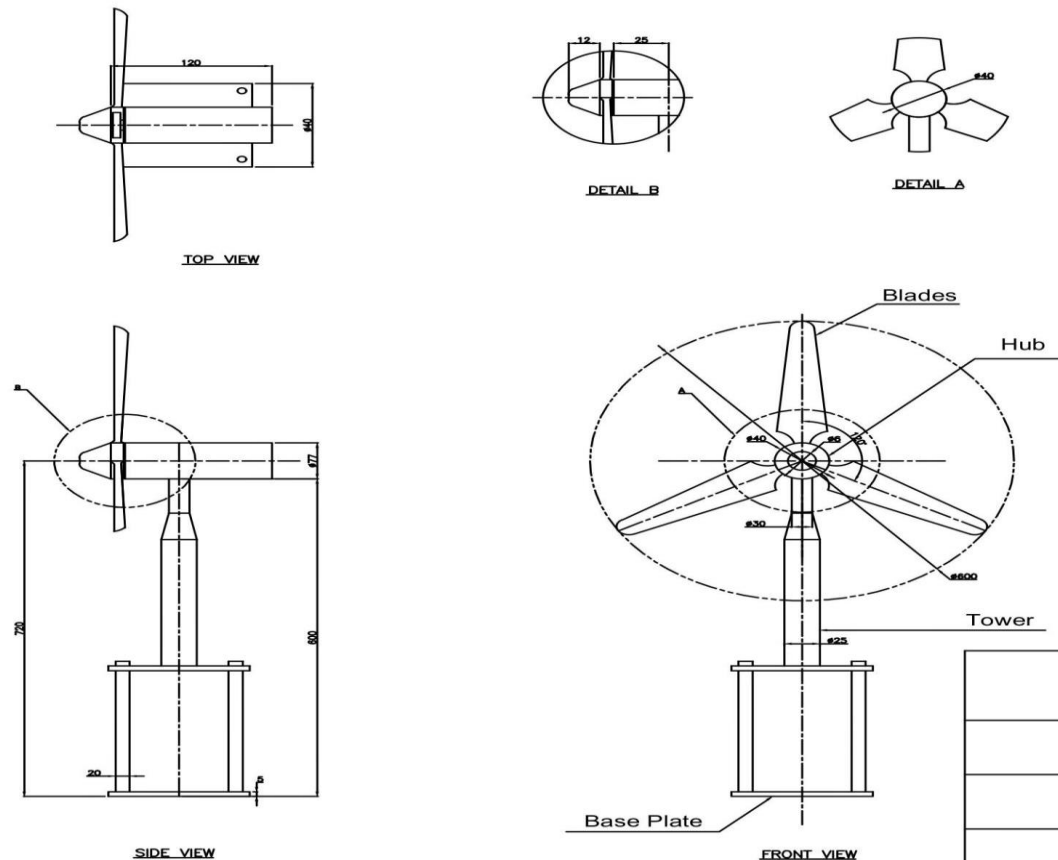
Functions of Components

- **Gearbox** :- The gearbox converts the rotor motion of lower rpm into the higher rpm.
- **Coupling**: Coupling is done between the main shaft and the transmission.
- **Generator**- Generates electricity from rotation of shaft.
- **Controller** :- It is a self operating system use in control system. It may be sensor (mechanical, electrical), decision elements(relays, microprocessor) and actuators(hydraulic, electric).
- **Yaw mechanism** :- Turns the nacelle with the rotor into the wind using motors
- **Anemometer and wind vane** :- Measures the speed and the direction of the wind while sending signals to the controller to start or stop the turbine



AutoCAD

AutoCAD is a commercial software for computer-aided design (CAD), used to create precise 2D and 3D drawings, models, and schematics.



S.No.	Component Name	Material Used	Qyt.
1.	Blades	PLA (3D Printer)	3
2.	Hub	PLA / Aluminum	1
3.	Main Shaft	Stainless Steel	1
4.	Bearings	PLA / Plastic	2
5.	Base Plate	Acrylic sheet	1
6.	Tower	PVC / Aluminum	1

Govt. Polytechnic Srinagar

Made By:-

All Dimensions In MM

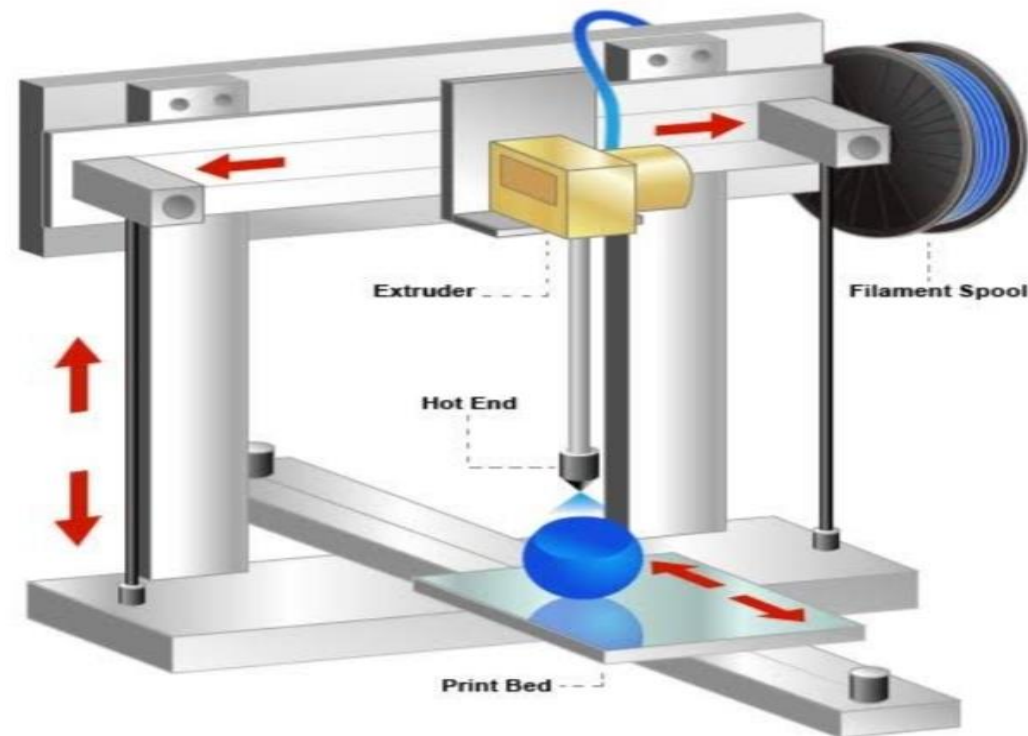
3D PRINTED PORTABLE
WIND TURBINE

Scale:-

2:1

3D printing

- 3D printing is a form of additive manufacturing technology where a three dimensional object is created by laying down successive layers of material.
- It is also known as Additive manufacturing.
- 3D printing is achieved using an additive process, where successive layers of material are laid down in different shapes



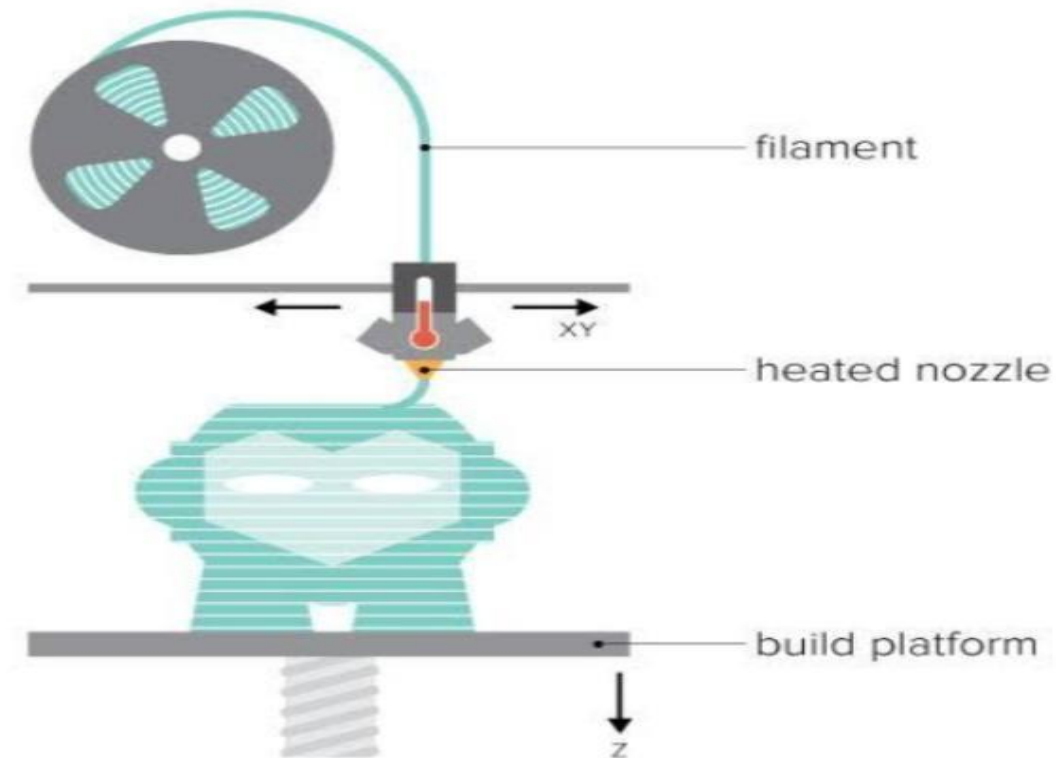
Working Principle

The operation of a 3D printing machine is based on the layer deposition of a thermoplastic material according to a digital model. A Computer-Aided Design (CAD) model is created first and then converted into a G-code file through slicing software. The printer's nozzle heats and extrudes the material in successive layers to form the final component. After printing, post-processing operations such as cleaning, sanding, or coating may be applied to improve the finish.



FDM Process is Used for Wind Turbine Components

The FDM process offers several advantages: it is cost-effective, easy to operate, and compatible with a variety of materials. It allows quick prototyping with a high degree of dimensional accuracy and minimal material wastage. Since the project focuses on a portable and lightweight turbine model, FDM is the most appropriate and economical choice.



FDM Process



Material Used: PLA (Polylactic Acid)

PLA, or Polylactic Acid, is a biodegradable thermoplastic derived from renewable resources such as corn starch or sugarcane. It is the most commonly used filament in FDM 3D printing due to its low printing temperature (190–210°C), smooth surface finish, and ease of handling. PLA is environmentally friendly, affordable, and suitable for lightweight structural applications.

Reasons for Using PLA Material

PLA has been selected for the fabrication of wind turbine parts due to its excellent printability and dimensional stability. It produces smooth airfoil surfaces, ensuring better aerodynamic performance of the blades. The low warping tendency of PLA ensures accurate blade geometry. Additionally, its lightweight nature makes it ideal for portable turbine models.



- **Advantages of Using 3D Printing in Wind Turbine Development**

1. Customizable blade geometry (NACA profile)
2. Rapid prototyping and easy modification
3. Low cost and minimal material waste
4. Lightweight and portable structure
5. Simplified assembly and repair

Conclusion

3D printing technology provides a rapid, efficient, and sustainable method for producing small-scale renewable energy prototypes. By using the FDM process with PLA material, a cost-effective and lightweight portable wind turbine can be developed. This approach demonstrates the integration of modern manufacturing with renewable energy applications, offering valuable insights for engineering design and innovation.

